AI ACADEMY · LEVEL 5 · GRADE 10 · WEEK 36 · WORKBOOK

Neural network experiment defense

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Image-pattern Network Investigation

THE CONNECTED WEEK

One complete experiment-defense record.

Keep claims, calculations, condition failures, data roles and missing evidence together.

Keep one habit: Recall → W1–W3/W5 → independent defense → final portfolio → delayed return.

MAIN LESSON · SECTION 1

Recall · Retrieve before notes

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

- Without opening the old lesson, explain one key idea from Capstone **limitation**s and revision. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Capstone hypothesis and baseline. Give an example and a mistake that the idea helps you avoid.

- Without opening the old lesson, explain one key idea from Dataset shift in neural models. Give an example and a mistake that the idea helps you avoid.

- Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

- A **defense** is not a sales pitch. Assemble the original question and criterion, data roles, baseline, model equation, settings, raw checkpoints, validation selection, frozen final **prediction**s, **limitation**s and revision history.

  Someone reading the record should be able to recompute a reported MSE and see which decisions each dataset informed.

- For this capstone, two rates were compared under matched conditions. Rate 0.2 selected a = 2 with validation MSE 0 versus 0.78125 for rate 0.1.

  The frozen candidate then achieved final MSE 5 versus baseline 20.25 on four constructed cases. These are different comparisons and must not be collapsed into “the network was 100% accurate”.

Capstone limitations and revision

> **My remembered idea, example and correction**

Capstone hypothesis and baseline

> **My remembered idea, example and correction**

Dataset shift in neural models

> **My remembered idea, example and correction**

Lesson reference for workbook practice

MAIN LESSON · SECTION 5

W1 · Explain the current example

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main capstone: ReLU(a*x), no bias, initial 1; train (1, 2), (2, 4), two full-batch updates. Rates .1/.2 end at a1.75/2, validation on (3, 6), (4, 8) has MSE .78125/0. Select .2 by lower validation MSE, then freeze a2. Final rows (5, 10), (6, 12), (−1, −2), (−2, −4): baseline a1 squared errors 25, 36, 4, 16, MSE 20.25; candidate 0, 0, 4, 16, MSE 5. Negative candidate subset MSE 10 over two rows.

  Improvement is relative to the initial-weight baseline, not a fitted signed line. Nonnegative output cannot represent negative targets. Proposed signed output requires a new recorded procedure and fresh reserved evaluation after development. No timing or energy measured. Separate supported comparison MSE 8→2 on four final rows: reduction 6 units or75% of baseline MSE, not classification accuracy.

  Separate picture neuron [2, 1] with weights [.5, −1], bias.2 gives z/output.2; second input 2 gives z−.8/output 0. No probability or accuracy measure is supplied for that neuron. Publicly inspected cases are teaching/exposed cases for the reader, not automatically fresh assessment evidence.

- Name eight items needed to check the capstone from question through final result.

> **My reasoning and evidence**

MAIN LESSON · SECTION 6

W2 · Trace the calculation

> Keep this task’s inputs,rule and response together.

Complete task and evidence

- Main capstone: ReLU(a*x), no bias, initial 1; train (1, 2), (2, 4), two full-batch updates. Rates .1/.2 end at a1.75/2, validation on (3, 6), (4, 8) has MSE .78125/0. Select .2 by lower validation MSE, then freeze a2. Final rows (5, 10), (6, 12), (−1, −2), (−2, −4): baseline a1 squared errors 25, 36, 4, 16, MSE 20.25; candidate 0, 0, 4, 16, MSE 5. Negative candidate subset MSE 10 over two rows.

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  Separate picture neuron [2, 1] with weights [.5, −1], bias.2 gives z/output.2; second input 2 gives z−.8/output 0. No probability or accuracy measure is supplied for that neuron. Publicly inspected cases are teaching/exposed cases for the reader, not automatically fresh assessment evidence.

- State the validation selection and frozen final comparison using their correct metrics and scopes.

> **My reasoning and evidence**

MAIN LESSON · SECTION 7

W3 · Repair a misconception

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main capstone: ReLU(a*x), no bias, initial 1; train (1, 2), (2, 4), two full-batch updates. Rates .1/.2 end at a1.75/2, validation on (3, 6), (4, 8) has MSE .78125/0. Select .2 by lower validation MSE, then freeze a2. Final rows (5, 10), (6, 12), (−1, −2), (−2, −4): baseline a1 squared errors 25, 36, 4, 16, MSE 20.25; candidate 0, 0, 4, 16, MSE 5. Negative candidate subset MSE 10 over two rows.

  Improvement is relative to the initial-weight baseline, not a fitted signed line. Nonnegative output cannot represent negative targets. Proposed signed output requires a new recorded procedure and fresh reserved evaluation after development. No timing or energy measured. Separate supported comparison MSE 8→2 on four final rows: reduction 6 units or75% of baseline MSE, not classification accuracy.

  Separate picture neuron [2, 1] with weights [.5, −1], bias.2 gives z/output.2; second input 2 gives z−.8/output 0. No probability or accuracy measure is supplied for that neuron. Publicly inspected cases are teaching/exposed cases for the reader, not automatically fresh assessment evidence.

- A presenter claims the experiment proves neural networks beat fitted lines and use less energy. What **evidence** is missing?

> **My reasoning and evidence**

MAIN LESSON · SECTION 9

W5 · Explain the limitation

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Main capstone: ReLU(a*x), no bias, initial 1; train (1, 2), (2, 4), two full-batch updates. Rates .1/.2 end at a1.75/2, validation on (3, 6), (4, 8) has MSE .78125/0. Select .2 by lower validation MSE, then freeze a2. Final rows (5, 10), (6, 12), (−1, −2), (−2, −4): baseline a1 squared errors 25, 36, 4, 16, MSE 20.25; candidate 0, 0, 4, 16, MSE 5. Negative candidate subset MSE 10 over two rows.

  Improvement is relative to the initial-weight baseline, not a fitted signed line. Nonnegative output cannot represent negative targets. Proposed signed output requires a new recorded procedure and fresh reserved evaluation after development. No timing or energy measured. Separate supported comparison MSE 8→2 on four final rows: reduction 6 units or75% of baseline MSE, not classification accuracy.

  Separate picture neuron [2, 1] with weights [.5, −1], bias.2 gives z/output.2; second input 2 gives z−.8/output 0. No probability or accuracy measure is supplied for that neuron. Publicly inspected cases are teaching/exposed cases for the reader, not automatically fresh assessment evidence.

- Propose a fair next comparison with a fitted signed line. State how to preserve previous **evidence** and reserve new evaluation.

> **My reasoning and evidence**

MAIN LESSON · SECTION 4

Try alone · Selected transfer

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The **written rule** selects lower MSE.

  After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

  Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

- Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

Independent case: Defend a new record independently

> **My independent explanation, trace and limits**

MAIN LESSON · SECTION 10

Project · Keep the mathematical evidence

> Keep this task's inputs,rule and response together.

Complete task and evidence

- For Image-pattern Network Investigation, use this week's independent task as a rehearsal for the final **evidence** pack: A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0.

  The **written rule** selects lower MSE. After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

  Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

  Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

> **My project change, supporting evidence and remaining limit**
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> | |
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MAIN LESSON · SECTION 11

Feedback · Repair and return

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference · Supported W2 record**

REFERENCE · SECTION 2

Reference · Supported W2 record

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use W2: Defend the result for Neural network experiment **defense**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

The step I completed without help, and the hint I needed

▼ Extra practice · Individual reasoning check

REFERENCE · SECTION 3

Extra practice · Individual reasoning check

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Use the worked case for Neural network experiment **defense**. Proposed explanation: "A polished demonstration is an evaluation". Decide whether this explanation is supported.

  Point to the step or supplied fact that decides; explain your repair if needed.

> **My explanation before partner discussion**

▼ Reference · Earlier W4 response copy

REFERENCE · SECTION 8

Reference · Earlier W4 response copy

> Keep this task's inputs,rule and response together.

Complete task and evidence

- A separate experiment compares candidates C1 and C2 on two validation rows: MSE 1.5 and 1.0. The **written rule** selects lower MSE.

  After freezing C2, four fresh final rows give baseline squared errors 4,9,1,2 and C2 squared errors 1,1,2,0. Compute final MSE and relative reduction.

  Two condition-B rows are the last two rows; compare baseline and candidate MSE there. State what this record supports and what is unmeasured.

> **My reasoning and evidence**

▼ **Extra practice · Pictured case**

REFERENCE · SECTION 12

Extra practice · Pictured case

> Keep this task's inputs,rule and response together.

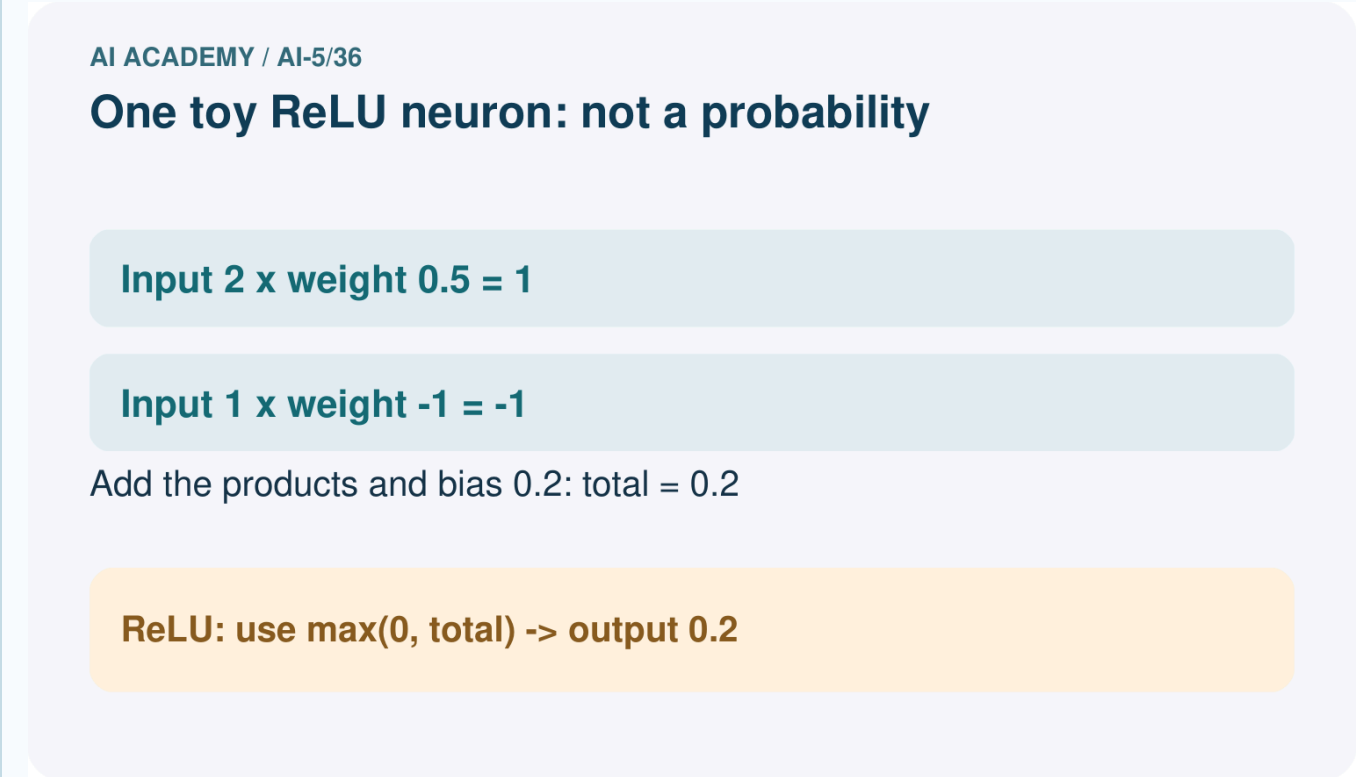
Complete task and evidence

- Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

- **Input**s are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

- Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.
- Support used: alone / hint / worked example. This record is practice **evidence**.
- Project connection: For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record.

  State exactly what this example supports before linking it to the wider capstone claim.



Follow the labelled picture with the matching explanation. Open the vector diagram

AI ACADEMY / AI-5/36

Neural network experiment defense

Start with the supplied facts

Follow each arrow. Ask which fact allows the next step.



1. Multiply inputs

2 x 0.5=1 and 1 x (-1)=-1.



2. Add weighted terms

Pre-activation z is 1-1+0.2=0.2.



3. Apply ReLU

max(0,0.2)=0.2; this activation is not a confidence percentage.

CHANGE ONE THING

Change the second input from 1 to 2.

Explain the changed result using the matching lesson.

Follow the labelled picture with the matching explanation. Open the vector diagram

My picture-based explanation and changed-case prediction

REFERENCE · SECTION 13

Reference · Project milestone

> Keep this task's inputs,rule and response together.

Complete task and evidence

- Project: Image-pattern Network Investigation
- Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.
- For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

▼ Extra practice · Predict,check and repair

REFERENCE · SECTION 14

Extra practice · Predict,check and repair

> Keep supported feedback separate from the independent case.

First supported question and optional hint

- Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.
- Which labelled step matches this detail? max(0,0.2)=0.2; this activation is not a confidence percentage.
- 1. Add weighted terms
- 2. Apply ReLU
- 3. Multiply **input**s
- Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Practice 1: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the first attempt**

After your attempt, compare with the supported explanation: Apply ReLU max(0,0.2)=0.2; this activation is not a confidence percentage.

A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

Changed-case question and optional hint

- Changed-case practice: Change the second **input** from 1 to 2. Which conclusion is supported by the supplied facts?
- 1. We can decide the result without examining the changed **input** or the procedure.
- 2. This one example guarantees correct results for every future **input**.
- 3. z becomes 1-2+0.2=-0.8 and ReLU returns 0.
- Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Practice 2: explain the deciding evidence

My selected conclusion, deciding fact and explanation

▼ **Check after the changed attempt**

After your attempt, compare with the supported explanation: z becomes 1-2+0.2=-0.8 and ReLU returns 0. The negative weighted term grows; ReLU clips a negative pre-activation to zero.

Repair and retain the record

Review Capstone **limitation**s and revision (ai-5/35). Goal: Evaluate a frozen candidate and baseline on final rows. Short practice: Calculate the slice rates and explain the claim revision.

If the team trains on the shifted evaluation patterns, what kind of **evidence** does their rerun provide?. Return to this week and explain what you changed.

Review Capstone hypothesis and baseline (ai-5/33). Goal: Write a testable capstone hypothesis and a credible baseline. Short practice: Name the intended change, the baseline and the measured outcome.

Why must both versions avoid **training** on the eight reserved labels?. Return to this week and explain what you changed.

Review Dataset shift in neural models (ai-5/30). Goal: Explain how a changed **input** mix can change overall accuracy. Short practice: Did the supplied model's dim-light accuracy deteriorate?

Point to the relevant step in the picture, then write the changed result and the rule or **evidence** that supports it.. Return to this week and explain what you changed.

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

▼ Earlier wording, alternatives and picture transcripts

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Repair a missing building block

Keep your completed work

▼ Reference · Six-picture response sheets

REFERENCE · SECTION 15

Reference · Six-picture response sheets

> Choose the picture that helps explain the current step.

01. Neural network experiment defense: the case

AI ACADEMY / AI-5/36 · 01 · LOOK AT THE FACTS

Neural network experiment defense: the case

The situation

Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

Supplied evidence

Inputs are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

Explain neural network experiment defense with a traceable toy-network case and a bounded conclusion.

Follow the labelled picture with the matching explanation. Open the vector diagram

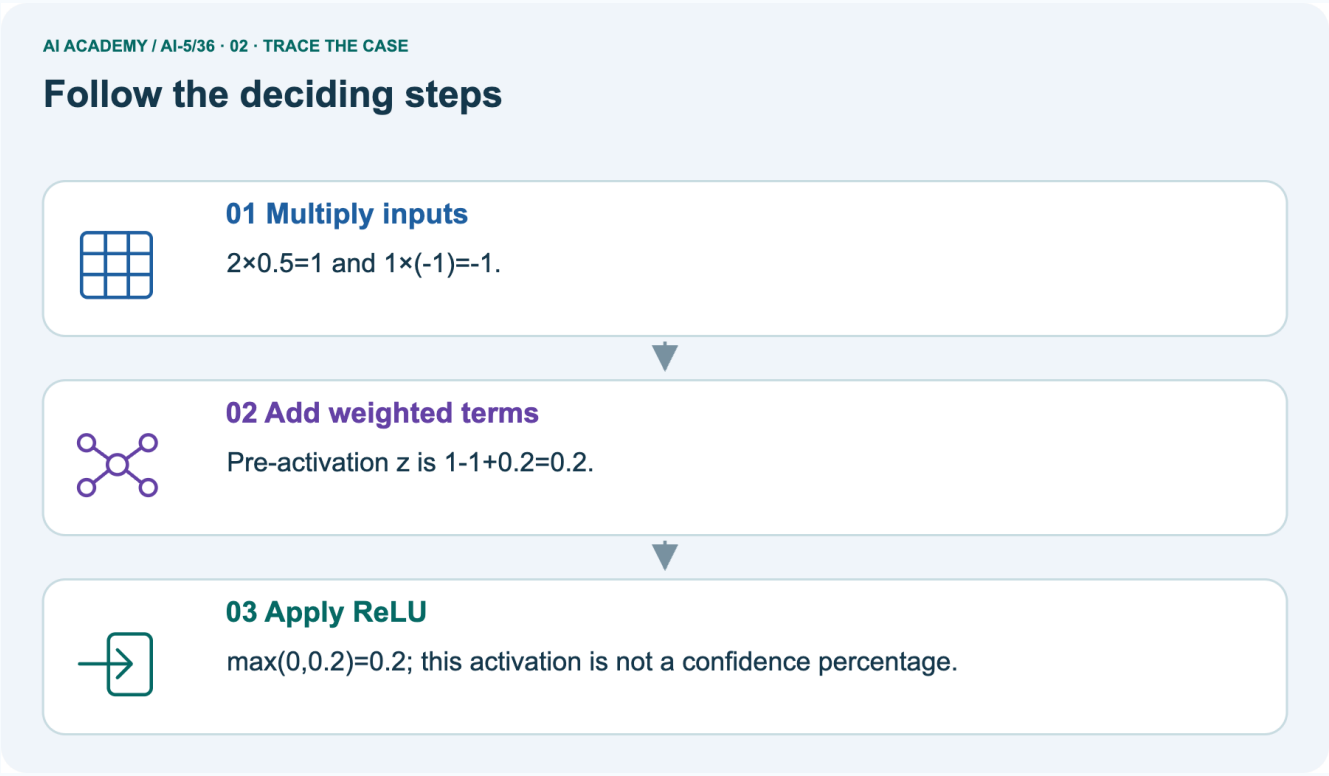
Defend an individual network calculation. At the defence, a team shows a single toy neuron before describing the whole image-pattern experiment.

Supplied facts: **Input**s are [2,1]; weights [0.5,-1]; bias 0.2; activation is ReLU, max(0,z). This is a demonstration, not a probability model.

Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



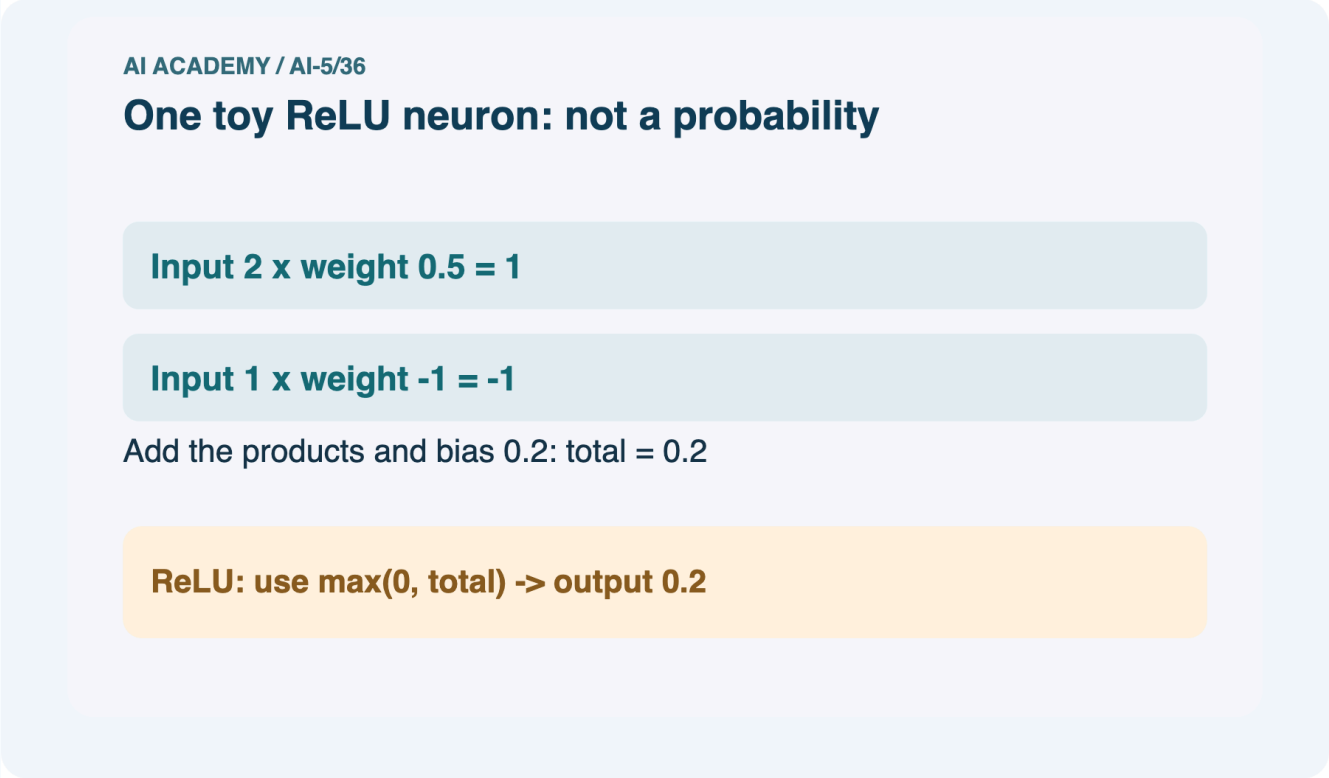
Follow the labelled picture with the matching explanation. Open the vector diagram

Multiply **input**s: 2×0.5=1 and 1×(-1)=-1. → Add weighted terms: Pre-activation z is 1-1+0.2=0.2. → Apply ReLU: max(0,0.2)=0.2; this activation is not a confidence percentage.

Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. One toy ReLU neuron: not a probability



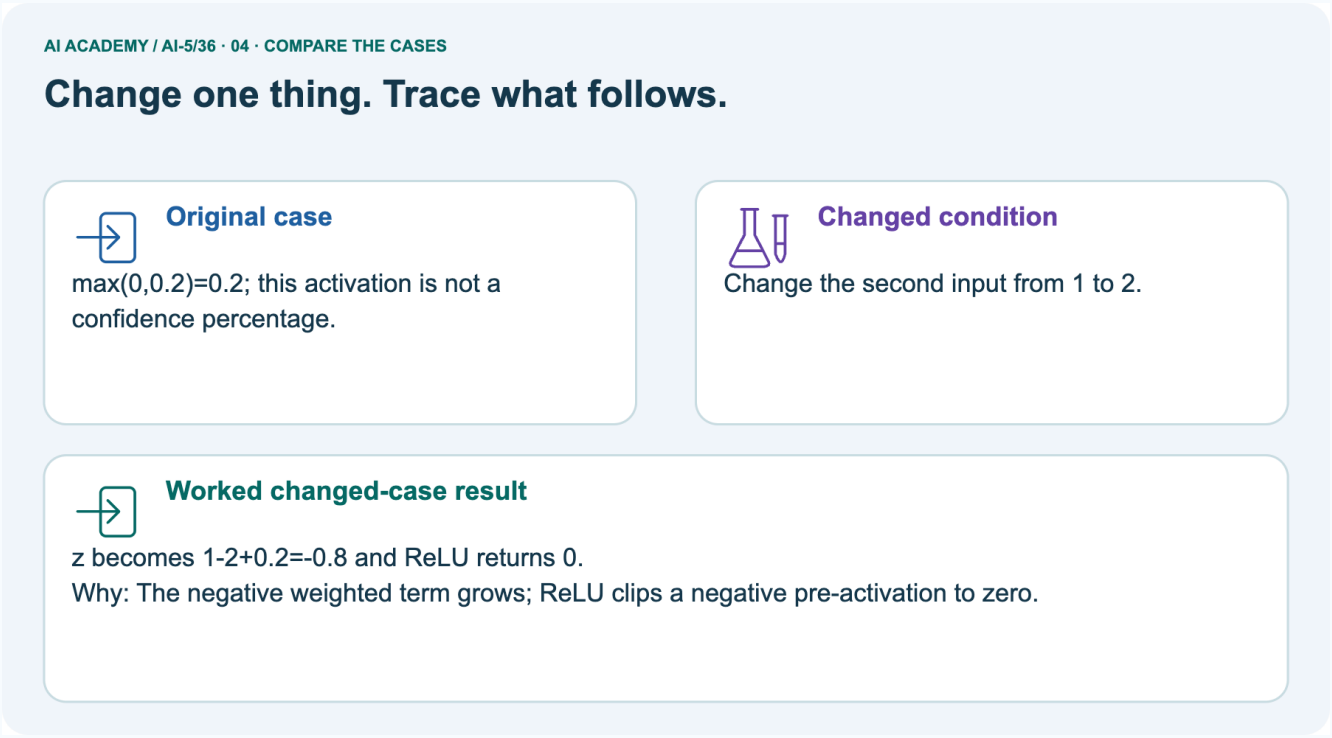
Follow the labelled picture with the matching explanation. Open the vector diagram

One toy ReLU neuron: not a probability. **Input**s: [2, 1]; weights: [0.5, -1]; bias: 0.2; activation: relu; **output**: 0.2.

Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

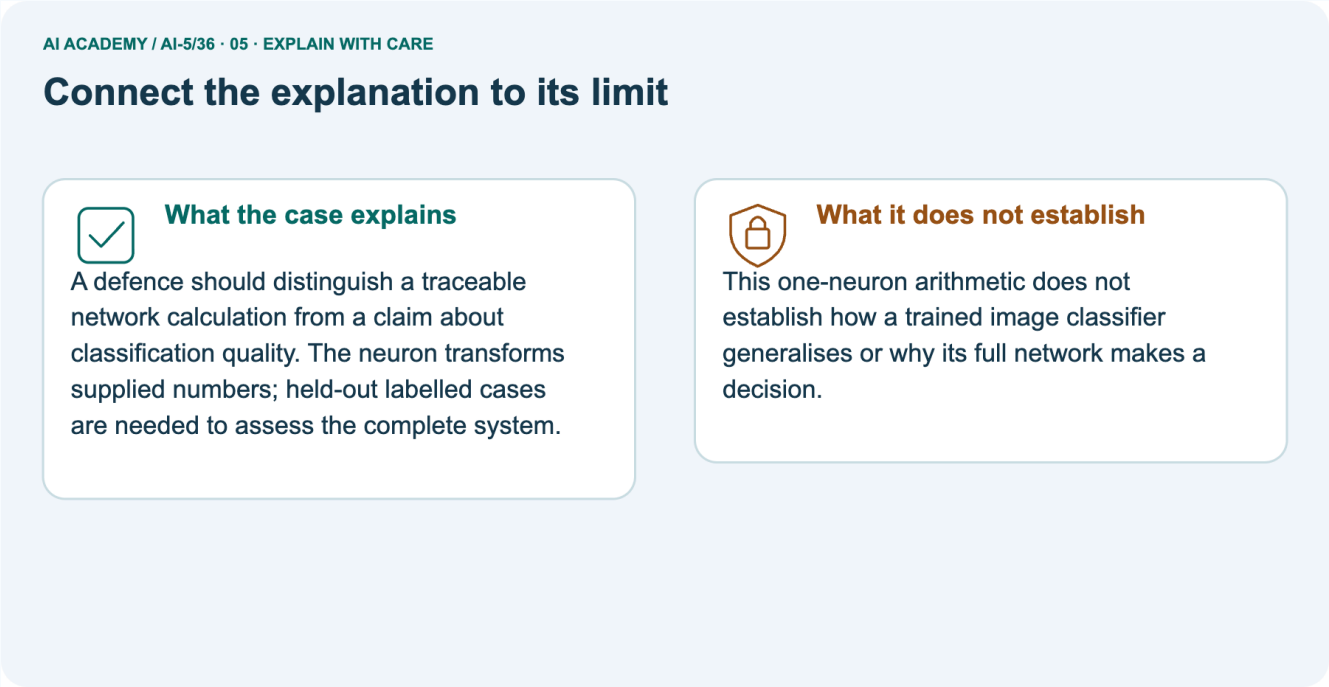
Original case: max(0,0.2)=0.2; this activation is not a confidence percentage. Changed condition: Change the second **input** from 1 to 2. Expected result: z becomes 1-2+0.2=-0.8 and ReLU returns 0.

Reason: The negative weighted term grows; ReLU clips a negative pre-activation to zero.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

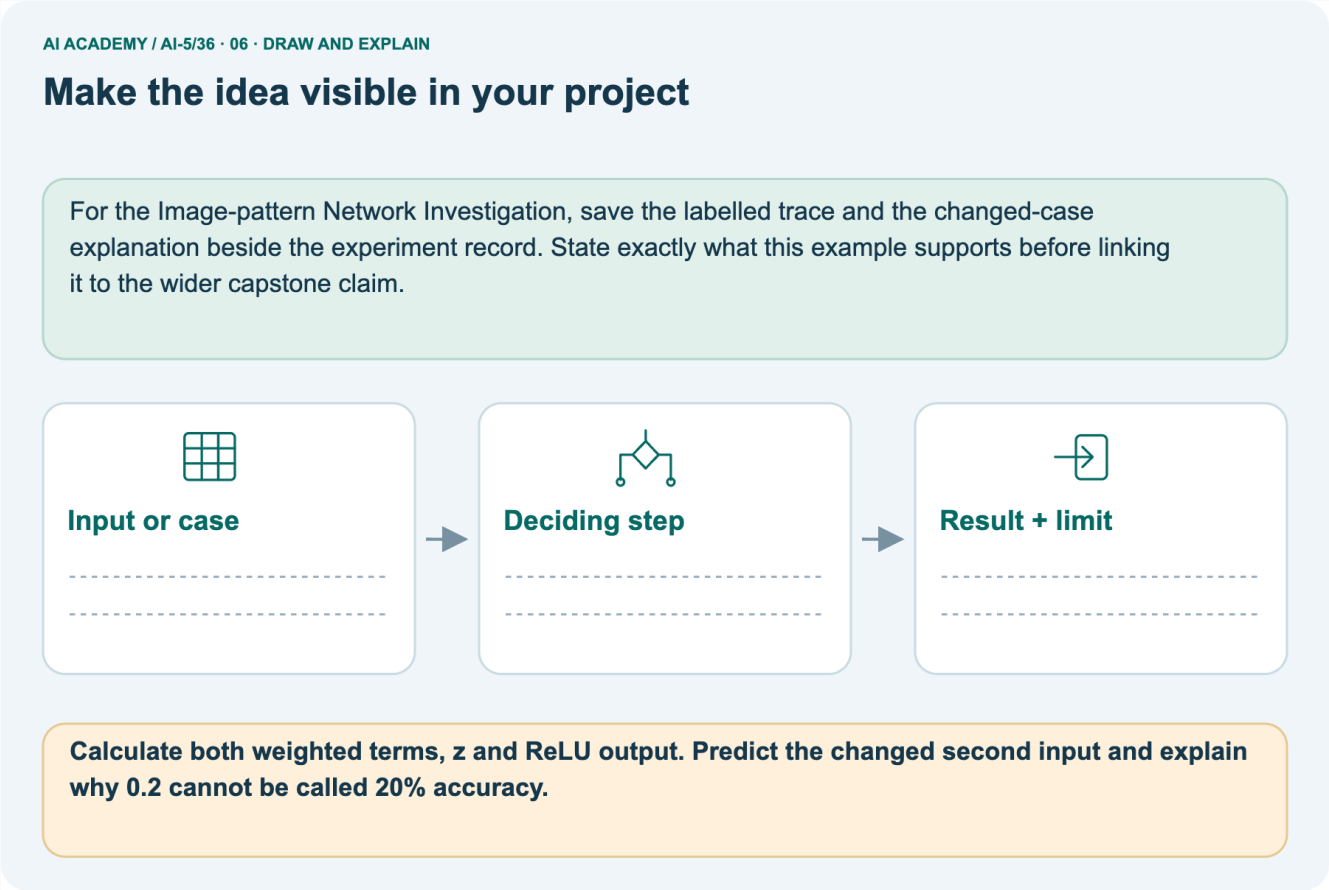
A defence should distinguish a traceable network calculation from a claim about classification quality. The neuron transforms supplied numbers; held-out labelled cases are needed to assess the complete system.

Limit: This one-neuron arithmetic does not establish how a trained image classifier generalises or why its full network makes a decision.

Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Image-pattern Network Investigation, save the labelled trace and the changed-case explanation beside the experiment record. State exactly what this example supports before linking it to the wider capstone claim.

Draw your **input**, deciding step, result and limit. Supported question: Calculate both weighted terms, z and ReLU **output**. Predict the changed second **input** and explain why 0.2 cannot be called 20% accuracy.

Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

▼ Earlier wording, alternatives and picture transcripts

01. Neural network experiment defense: the case

02. Follow the deciding steps

03. One toy ReLU neuron: not a probability

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.